

MEDREPORTAGENT: AN AI-POWERED SINGLE-AGENT LARGE LANGUAGE MODEL SYSTEM FOR AUTOMATED MEDICAL LABORATORY REPORT EXPLANATION

Muneer Hussain & Syed Ismail

Department of Software Engineering
University of Engineering and Technology
Taxila, Pakistan
22-se-71@students.uettaxila.edu.pk

Dr.Kanwal Yousaf

Department of Software Engineering
University of Engineering and Technology
Taxila, Pakistan
kanwal.yousaf@uettaxila.edu.pk

ABSTRACT

Advancements in large language models (LLMs) have opened new possibilities for building intelligent medical assistance tools that can interpret complex clinical data using human-friendly interfaces. However, existing systems often lack domain-specific reasoning, contextual interpretation, and accessible output formatting essential for patient-facing applications. We present MedReportAgent, a single-agent LLM-based system powered by the LLaMA 3.3 70B model via the Groq API, designed to automatically analyze and explain medical laboratory reports in simple, non-technical language. The system accepts both textual lab values and PDF-based reports as inputs, processes them through a carefully engineered system prompt defining structured analytical roles, and returns value-by-value interpretations including status classification (HIGH, NORMAL, LOW), reference range comparison, and organ-system mapping. Furthermore, MedReportAgent integrates a Flask-based web interface with a real-time chat functionality and automated PDF report generation via ReportLab.

We evaluated our system on a benchmark set of 50 synthetic and de-identified real medical laboratory reports covering blood chemistry, liver function, thyroid panel, lipid profile, and complete blood count domains. MedReportAgent achieved a structured output compliance rate of 96%, a reference-range accuracy of 92.4%, and a patient comprehensibility score of 4.3/5.0 based on user study feedback. These results demonstrate that a well-engineered single-agent LLM system with domain-specific prompting can serve as a reliable, interpretable, and accessible medical report explanation tool. Our approach provides transparent, step-by-step explanations for each laboratory value — a critical feature for patient education and healthcare accessibility in resource-limited settings. Code is available at [<https://github.com/MuneerHussain01/AI-Based-Medical-Report-Explanation-Agent>].

1 INTRODUCTION

Large Language Models (LLMs) have demonstrated remarkable capabilities in solving a wide range of problems using human-friendly interfaces (Wei et al., 2022a). The healthcare domain, in particular, presents a compelling use case for LLM-based agents, given the complexity of medical documentation and the significant gap between clinical terminology and patient comprehension. Medical laboratory reports — containing values for blood counts, biochemical panels, hormonal assays, and other diagnostic measurements — are routinely issued to patients who often lack the domain knowledge to interpret them meaningfully. Misinterpretation or ignorance of abnormal laboratory findings can delay critical healthcare decisions, leading to adverse patient outcomes (Patel et al., 2021).

To address these challenges, there is growing interest in AI-driven tools that can bridge the communication gap between laboratory diagnostics and patient understanding. Recent advances in retrieval-augmented generation (RAG) (Lewis et al., 2020), chain-of-thought prompting (Wei et al., 2022b), and instruction-following LLMs have made it feasible to build specialized medical AI agents that can process unstructured clinical text and produce structured, interpretable

outputs. However, most existing systems either focus on clinical decision support for medical professionals or require complex multi-agent orchestration frameworks that impose significant computational and infrastructure overhead.

In this paper, we propose MedReportAgent, a single-agent LLM system designed specifically for the task of medical laboratory report explanation. Our system is built on the LLaMA 3.3 70B model, accessed through the Groq API for low-latency inference, and is deployed via a Flask-based web server with an integrated real-time chat interface. The key design philosophy of MedReportAgent is domain-specific prompt engineering — a carefully structured system prompt that defines the agent's analytical responsibilities, output format, and safety constraints. This approach eliminates the need for multi-agent orchestration while still achieving high-quality, structured medical explanations.

The primary contributions of this paper are as follows:

- We present MedReportAgent, an end-to-end medical report explanation system using a single LLM agent with structured prompt engineering.
- We introduce a domain-specific system prompt architecture that enforces structured output including status classification, reference range comparison, and organ-system mapping.
- We integrate PDF input processing (via PyPDF2) and automated PDF report generation (via ReportLab) into a unified Flask web application.
- We evaluate the system on 50 medical laboratory report cases and demonstrate high accuracy in reference range classification, structured output compliance, and patient comprehensibility.

2 RELATED WORKS

Large Language Models in Healthcare

LLMs have been increasingly applied to healthcare tasks, ranging from clinical note summarization (Patel et al., 2023) to medical question answering (Singhal et al., 2023). Models such as GPT-4 (OpenAI, 2023) and Med-PaLM 2 (Singhal et al., 2023) have demonstrated near-expert performance on medical licensing examinations. However, these systems are generally designed for clinical professional audiences and do not optimize for patient-accessible explanations of routine diagnostic data.

Medical Report Analysis

Automated analysis of medical reports has been explored using rule-based systems (Friedman et al., 2004), named entity recognition (Uzuner et al., 2011), and more recently, transformer-based architectures (Lee et al., 2020). BioBERT (Lee et al., 2020) and ClinicalBERT (Huang et al., 2019) demonstrated the utility of domain-adapted language models for clinical text processing. Our approach differs in that we leverage a general-purpose instruction-following LLM with domain-specific prompt engineering rather than domain-adapted model fine-tuning, enabling rapid deployment without specialized training data.

LLMs with Structured Reasoning

Chain-of-Thought (CoT) prompting (Wei et al., 2022b) enables LLMs to produce intermediate reasoning steps, improving performance on complex analytical tasks. Instruction-following techniques (Ouyang et al., 2022) further allow models to adhere to structured output formats, which is critical for medical applications requiring consistent and auditable outputs. MedReportAgent incorporates these principles by embedding explicit analytical steps and output format constraints within the system prompt.

AI-Powered Medical Chatbots

Conversational AI systems for healthcare, such as Babylon Health (Babylon Health, 2020) and ADA Health (ADA Health, 2021), have demonstrated the viability of symptom-checking chatbots for patient triage. However, these systems do not specifically address the interpretation of quantitative laboratory values against clinical reference ranges.

MedReportAgent fills this gap by providing value-by-value laboratory report analysis within a conversational interface, supporting both text input and PDF document upload.

Groq and Low-Latency LLM Inference

The Groq Language Processing Unit (LPU) architecture provides significantly faster token generation compared to traditional GPU-based inference (Groq, 2024), making it suitable for real-time conversational applications. By routing LLaMA 3.3 70B through the Groq API, MedReportAgent achieves low-latency responses essential for an interactive medical explanation experience.

3 METHODS

3.1 Overview of MedReportAgent

MedReportAgent is a single-agent LLM system designed for the automated explanation of medical laboratory reports. Unlike multi-agent architectures that distribute analytical tasks across specialized sub-agents, MedReportAgent concentrates all analytical responsibilities within a single, comprehensively prompted LLM instance. This design choice is motivated by the well-defined and structured nature of laboratory report interpretation, which does not require the diversity of evidence sources characteristic of open-domain scientific queries such as drug-target interaction prediction.

The system workflow proceeds as follows:

1. The user submits a laboratory report either as free-text input via the chat interface or as a PDF document uploaded through the web interface.
2. For PDF inputs, the system extracts text using the PyPDF2 library and constructs a structured query message containing the extracted laboratory values.
3. The query is forwarded to the LLaMA 3.3 70B model via the Groq API, accompanied by a fixed system prompt that defines the agent's role, analytical process, output format, and safety constraints.
4. The model generates a structured response following the defined output format, which is parsed and rendered in the web interface with visual status indicators (HIGH, NORMAL, LOW tags).
5. Optionally, the user may download the analysis as a formatted PDF report generated by the ReportLab library.

All interactions are managed by a Flask web server exposing three primary API endpoints: `/api/analyze` for text-based queries, `/api/analyze-pdf` for document-based queries, and `/api/generate-pdf` for report download. The system maintains a client-side conversation history to support multi-turn interactions within a single session.

3.2 Agent Role and Responsibilities

The core intelligence of MedReportAgent is concentrated in a single LLM agent whose behavior is governed entirely by the system prompt. The agent is instructed to perform the following analytical responsibilities for each medical laboratory report submitted:

- Analyze each laboratory value (blood tests, urine tests, liver function, thyroid, lipid profile, complete blood count, etc.) against standard clinical reference ranges.
- Classify each value as HIGH, NORMAL, or LOW based on established reference interval data.
- Explain each result in simple, non-technical English accessible to patients without medical training.
- Identify the organ or body system associated with each laboratory test.
- Enumerate possible general causes of abnormal values without providing a definitive diagnosis.
- Provide actionable general wellness tips relevant to the reported findings.
- Specify warning signs from the report that warrant prompt medical consultation.

3.3 System Prompt Architecture

The system prompt is the primary mechanism for constraining and directing the LLM agent's behavior. MedReportAgent employs a structured system prompt with three major components: Role Definition, Output Format Specification, and Safety Rules.

The Role Definition establishes the agent as an expert medical report explanation AI designed to help patients understand their laboratory results. The Output Format Specification enforces a consistent five-section structure for all responses: (1) Report Summary, (2) Value-by-Value Analysis, (3) Key Observations, (4) General Health Tips, and (5) When to See a Doctor. Each section is defined with explicit content requirements to ensure completeness and consistency across responses. The Safety Rules prohibit the agent from providing definitive diagnoses, recommending specific medications, or generating content that could substitute for professional medical consultation.

The output structure for the Value-by-Value Analysis section follows the format:

```
Test Name: [value] [unit]    Status: HIGH / NORMAL / LOW    Normal Range:
[range]    What it means: [explanation]    Related to: [organ/system]
```

This structured format enables reliable downstream parsing for visual rendering in the web interface and for automated PDF report generation.

3.4 LLM Backbone and API Configuration

MedReportAgent utilizes the LLaMA 3.3 70B Versatile model (Meta AI, 2024) as its backbone LLM, accessed through the Groq API (Groq, 2024). LLaMA 3.3 70B was selected for its strong performance on instruction-following tasks, its open-weight availability enabling deployment flexibility, and its demonstrated competency in biomedical text processing. The model is queried with a temperature setting of 0.3 to balance response determinism with linguistic naturalness, and a maximum token budget of 2,048 tokens per response to accommodate comprehensive laboratory panel analyses.

The Groq API provides hardware-accelerated inference via its custom Language Processing Unit (LPU) architecture, achieving token generation speeds significantly exceeding standard GPU-based deployments. This low-latency characteristic is critical for maintaining a responsive conversational user experience in the MedReportAgent web interface.

3.5 PDF Processing Pipeline

MedReportAgent supports two input modalities: plain text and PDF documents. For PDF inputs, the system implements a two-stage processing pipeline. In the extraction stage, the PyPDF2 library is used to iterate over all pages of the uploaded document and concatenate extracted text content. Minimum text length validation (20 characters) is applied to reject image-based or encrypted PDFs that yield insufficient extractable content. In the query construction stage, the extracted text is embedded into a structured prompt instructing the model to analyze the contained laboratory report values.

For PDF output generation, MedReportAgent uses the ReportLab library to construct formatted PDF documents from the model's plain-text analysis output. The generation pipeline parses section headers (##), subsection headers (###), bullet points (-), and body text from the markdown-formatted model response, applying corresponding typographic styles including heading formats, bullet indentation, and horizontal rule separators. Non-Latin characters and emoji symbols are sanitized prior to PDF generation to ensure compatibility with ReportLab's rendering engine.

3.6 Web Interface Architecture

The MedReportAgent user interface is implemented as a single-page web application served by the Flask framework. The interface provides a real-time chat environment supporting both text and PDF file inputs. Key interface components include: an input text area with keyboard shortcut support (Enter to send, Shift+Enter for newline), a PDF file upload button with client-side base64 encoding, a typing indicator displayed during model inference, color-coded status tags

(HIGH in red, LOW in blue, NORMAL in green) applied via client-side response formatting, and a per-response PDF download button that triggers the `/api/generate-pdf` endpoint. Chat history is maintained client-side across the session to support contextually coherent multi-turn interactions.

4 EXPERIMENT

4.1 Dataset and Evaluation Setup

To evaluate MedReportAgent, we constructed a benchmark dataset of 50 laboratory report cases. The dataset comprises 30 synthetic laboratory reports generated to cover a range of abnormal and normal value combinations across six common panel types — Complete Blood Count (CBC), Basic Metabolic Panel (BMP), Liver Function Tests (LFT), Lipid Profile, Thyroid Function Tests (TFT), and Urinalysis — and 20 de-identified anonymized real laboratory report excerpts sourced from publicly available medical education repositories. All reports were reviewed for clinical plausibility by a medical professional prior to evaluation.

We evaluated MedReportAgent on three primary metrics: (1) Structured Output Compliance Rate — the proportion of responses fully adhering to the five-section output format specified in the system prompt; (2) Reference Range Classification Accuracy — the proportion of individual laboratory values correctly classified as HIGH, NORMAL, or LOW relative to established clinical reference intervals; and (3) Patient Comprehensibility Score — a subjective rating (1–5 scale) collected from 15 non-medical participants who rated the clarity and usefulness of the explanations generated for a randomly selected subset of 20 reports.

4.2 Performance Results

Table 1 summarizes the evaluation results for MedReportAgent across all three metrics, alongside a comparison with two baseline configurations: a zero-shot GPT-4o mini query without a structured system prompt (Baseline-A), and a rule-based reference range lookup system with templated output (Baseline-B).

Table 1: Evaluation results across models. Bold indicates best performance.

Metric	MedReportAgent	Baseline-A (GPT-4o mini, no prompt)	Baseline-B (Rule-based)	Better
Output Compliance Rate	96.0%	54.0%*	88.0%*	↑
Reference Range Accuracy	92.4%	71.2%*	95.1%	↑
Comprehensibility Score (1-5)	4.3 / 5.0	2.9 / 5.0*	3.1 / 5.0*	↑
Avg. Response Latency	~3.2s	~8.5s	~0.4s	↓
PDF Input Support	Yes	No	No	-
PDF Report Generation	Yes	No	No	-

Results show means over 50 test cases. * denotes statistical significance ($p\text{-value} < 0.05$) compared to MedReportAgent. Arrows (↑/↓) indicate better direction.

MedReportAgent demonstrates consistently strong performance across all evaluation dimensions. The 96.0% structured output compliance rate reflects the effectiveness of explicit output format specification in the system prompt, far exceeding the 54.0% rate of the unstructured baseline (Baseline-A). While Baseline-B achieves slightly higher reference range accuracy (95.1% vs. 92.4%) due to direct lookup against a curated reference interval database, MedReportAgent provides substantially richer contextual explanations that account for the patient comprehensibility score advantage (4.3 vs. 3.1). The 3.2-second average response latency, enabled by Groq’s LPU acceleration, represents a practical balance between response quality and user experience requirements for a conversational medical assistant.

4.3 Case Study Example

To illustrate MedReportAgent's operation, we present an example case involving a blood chemistry panel with multiple abnormal values. The user submits the following laboratory values as a text query: Glucose: 145 mg/dL, HbA1c: 7.2%, Creatinine: 1.9 mg/dL, eGFR: 42 mL/min/1.73m², ALT: 22 U/L, Total Cholesterol: 218 mg/dL.

MedReportAgent produces the following structured analysis excerpt:

Report Summary: This panel indicates elevated blood sugar, compromised kidney function, and borderline cholesterol levels. Immediate medical consultation is strongly recommended.

Glucose: 145 mg/dL — Status: HIGH | Normal Range: 70-99 mg/dL | Related to: Pancreas / Endocrine System | Possible causes: Diet, diabetes, stress response.

eGFR: 42 mL/min/1.73m² — Status: LOW | Normal Range: ≥60 | Related to: Kidneys | This suggests reduced kidney filtration capacity. Please consult a nephrologist promptly.

This example demonstrates MedReportAgent's ability to integrate clinical knowledge, patient-accessible language, and structured formatting within a single model inference cycle, without requiring external knowledge graph queries or multi-agent coordination.

5 DISCUSSION

Our study presents MedReportAgent, a single-agent LLM system for automated medical laboratory report explanation. The system demonstrates that carefully engineered system prompt architecture can substitute for multi-agent orchestration in domains with well-structured analytical tasks. By consolidating all analytical responsibilities — reference range classification, organ-system mapping, cause enumeration, and wellness guidance — within a single instruction-following LLM, MedReportAgent achieves high output quality without the architectural complexity, latency overhead, and inter-agent communication costs characteristic of multi-agent frameworks.

A key strength of MedReportAgent is its dual-modality input support. The integration of PDF text extraction enables users to directly upload their physical laboratory reports rather than manually transcribing individual values, significantly reducing the barrier to use. Combined with automated PDF report generation, the system provides a complete end-to-end workflow from report submission to downloadable patient-readable explanation document.

The system's safety architecture is a critical design consideration for medical AI applications. The system prompt explicitly prohibits definitive disease diagnosis and medication recommendation, ensuring that MedReportAgent functions as a patient education tool rather than a clinical decision support system. This boundary is reinforced by the mandatory inclusion of a disclaimer in every response, directing users to consult qualified medical professionals for diagnosis and treatment decisions.

However, several limitations exist in the current system. First, MedReportAgent's reference range classifications rely on the LLM's internally encoded clinical knowledge, which may not reflect the most current or population-specific reference interval standards. Integrating a curated, regularly updated reference interval database as a retrieval-augmented knowledge source would improve classification accuracy. Second, the system does not currently support image-based PDF reports, which constitute a significant proportion of real-world laboratory documents. Future work should investigate OCR integration (e.g., Tesseract, AWS Textract) to expand input coverage. Third, the system lacks support for multi-language outputs, limiting accessibility for non-English-speaking patient populations. Finally, the system prompt's effectiveness depends on the instruction-following capabilities of the backbone LLM; performance may degrade with smaller, less capable models deployed in resource-constrained settings.

Future work should focus on: (1) integrating a structured clinical reference interval database for enhanced classification accuracy, (2) adding OCR-based processing for image-based PDF reports, (3) multi-language output support for global accessibility, (4) longitudinal report comparison to identify trends across multiple laboratory visits, and (5) a prospective clinical validation study assessing real-world patient outcomes from use of the system.

6 FUNDING

This research was conducted as part of an academic course project at University of Engineering and Technology, Taxila. No external funding was received for this work.

REFERENCES

- ADA Health. (2021). ADA: Your health companion. ADA Health GmbH. <https://ada.com>
- Babylon Health. (2020). Babylon: Digital-first healthcare. Babylon Holdings Ltd. <https://www.babylonhealth.com>
- Friedman, C., Shagina, L., Lussier, Y., & Hripcsak, G. (2004). Automated encoding of clinical documents based on natural language processing. *Journal of the American Medical Informatics Association*, 11(5), 392-402.
- Groq. (2024). Groq language processing unit: Architecture and performance. Groq Inc. <https://groq.com>
- Huang, K., Altosaar, J., & Ranganath, R. (2019). ClinicalBERT: Modeling clinical notes and predicting hospital readmission. *arXiv preprint arXiv:1904.05342*.
- Lee, J., Yoon, W., Kim, S., Kim, D., Kim, S., So, C. H., & Kang, J. (2020). BioBERT: A pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics*, 36(4), 1234-1240.
- Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., ... & Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. *Advances in Neural Information Processing Systems*, 33, 9459-9474.
- Meta AI. (2024). LLaMA 3.3 Technical Report. Meta Platforms, Inc. <https://ai.meta.com>
- OpenAI. (2023). GPT-4 Technical Report. *arXiv preprint arXiv:2303.08774*.
- Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., ... & Lowe, R. (2022). Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 35, 27730-27744.
- Patel, V. L., Shortliffe, E. H., Stefanelli, M., Szolovits, P., Berthold, M. R., Bellazzi, R., & Abu-Hanna, A. (2021). The coming of age of artificial intelligence in medicine. *Artificial Intelligence in Medicine*, 46(1), 5-17.
- Patel, S. B., Lam, K., & Patel, B. (2023). ChatGPT: The future of discharge summaries? *The Lancet Digital Health*, 5(3), e107-e108.
- Singhal, K., Azizi, S., Tu, T., Mahdavi, S. S., Wei, J., Chung, H. W., ... & Natarajan, V. (2023). Large language models encode clinical knowledge. *Nature*, 620(7972), 172-180.
- Uzuner, O., South, B. R., Shen, S., & DuVall, S. L. (2011). 2010 i2b2/VA challenge on concepts, assertions, and relations in clinical text. *Journal of the American Medical Informatics Association*, 18(5), 552-556.
- Wei, J., Tay, Y., Bommasani, R., Raffel, C., Zoph, B., Borgeaud, S., ... & Fedus, W. (2022a). Emergent abilities of large language models. *arXiv preprint arXiv:2206.07682*.
- Wei, J., Wang, X., Schuurmans, D., Bosma, M., Xia, F., Chi, E., ... & Zhou, D. (2022b). Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems*, 35, 24824-24837.

A IMPLEMENTATION DETAILS

A.1 System Prompt (Full)

The complete system prompt used by MedReportAgent is reproduced below. This prompt is passed as the system role message to every API call and remains fixed across all user interactions:

"You are an expert Medical Report Explanation AI Agent designed to help patients understand their medical laboratory reports. Your responsibilities: (1) Analyze medical lab values against standard reference ranges, (2) Clearly mark each result as HIGH, NORMAL, or LOW, (3) Explain in simple non-technical English, (4) Identify organ/body system for each test, (5) Mention possible general causes of abnormal values without diagnosing, (6) Provide general wellness tips, (7) Always strongly recommend consulting a qualified doctor. OUTPUT FORMAT: Report Summary / Value-by-Value Analysis / Key Observations / General Health Tips / When to See a Doctor. RULES: Never diagnose, never prescribe, use simple English, be compassionate, urge immediate attention for critically abnormal values."

A.2 Software and Hardware Configuration

MedReportAgent was developed and tested on a standard laptop environment with Python 3.10. Key software dependencies include Flask 2.3.x for web server functionality, PyPDF2 3.0.x for PDF text extraction, ReportLab 4.x for PDF report generation, and the Requests library for Groq API communication. The LLM backend utilizes LLaMA 3.3 70B Versatile via the Groq API with hardware-accelerated inference. No specialized GPU hardware is required on the client side.

A.3 Example Use Case

To illustrate a complete system interaction, consider a user submitting the following query:

User Input: "My blood test shows: Hemoglobin: 9.2 g/dL, WBC: 11,500/uL, Platelets: 420,000/uL, Ferritin: 6 ng/mL. Can you explain these results?"

MedReportAgent processes this input through the Groq API and returns a structured five-section analysis. Hemoglobin is classified as LOW (normal: 12.0-17.5 g/dL), attributed to the hematopoietic system, with an explanation noting that this level may indicate anemia and listing possible causes such as iron deficiency, chronic disease, or nutritional deficiencies. WBC is classified as HIGH (normal: 4,500-11,000/uL), attributed to the immune system, noting possible causes such as infection or inflammation. Ferritin at 6 ng/mL is classified as LOW (normal: 12-300 ng/mL for females, 12-400 ng/mL for males), indicating depleted iron stores. The response concludes with a recommendation to consult a hematologist or primary care physician for further evaluation and iron supplementation assessment. A PDF download of this complete analysis is available to the user immediately upon response rendering.